ALEXANDER TONG

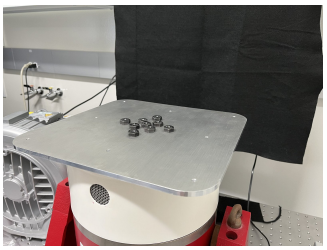
Mechanical engineering | Recovery hardware and supersonic aerodynamics

Boston University, B.S. Mechanical Engineering, Aerospace Concentration | Expected May 2028
alto335@bu.edu | 860-706-3624 | alexandertong.space

PORTFOLIO AT A GLANCE

Project Index

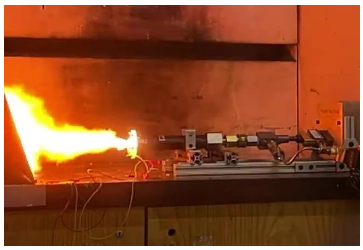
Page 2 / Separation & recovery	Earlier black-powder configuration: 18.7 lbm within a 20 lbm allocation. Load sizing, CO2 redesign and an operational vacuum chamber.
Page 3 / Aerodynamic fairing	Three concept studies; Ansys CFD baseline converged and under review. Shared aerodynamic analysis, integration and manufacturing trade.
Page 4 / Plummer shaker hardware	Structural analysis, bored-bolt checks and CNC fabrication. Installed plate-and-bolt hardware at 93.9% lower estimated cost.



Plummer: machined adapter installed.



AIAA: assembled Prototype 1 aircraft.



Hybrid nozzle: hotfire test frame.

Additional Projects

Plummer Laboratory

CNC-machined shaker adapter; plate-and-bolt hardware estimate reduced from $1,170 to $71.11, excluding labor. Designed the matrix and procedure for 48 Instron tests.

AIAA Design-Build-Fly wing

Complete Prototype 1 wing 42.7% lighter than the previous complete wing; build time reduced from three months to one. Led design, fabrication and member training.

Hybrid rocket nozzle

73% mass reduction from my initial design; approximately two-second hotfire. Minimum yield FOS 7.8 in SolidWorks FEA for mechanical loads only; thermal effects excluded. Thrust gain unmeasured.

EK210 line-following car

Designed and programmed the vehicle: six loaded course runs in 70-86 seconds against a 90-second limit. Recorded component cost $73.94 against a $200 budget.

Culturon

Executed the company's first full-scale nanoparticle deposition map using microscopy and ImageJ. Designed adhesive-free sample racks; presented prevention concepts.

Manufacturing & mentoring

CNC and manual machining, hot-wire foam cutting and composite fabrication. Taught BURPG members turning and milling through the machinist's-hammer project beginning Fall 2025.

Additional research & coursework

Space Physics & Technology Lab: instrument-data analysis with Python/pySPEDAS and assistance with flight-spare laser calibration. University of Sydney pulsed plasma thruster: inductor-spool design and coil-winding calculations. ME360 course project includes completed linear-stage/CNC drawing and fishing-game builds.

01 / RECOVERY HARDWARE

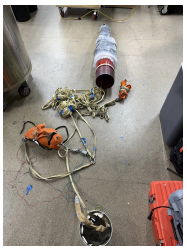
Argo Separation and Recovery Subsystem

Boston University Rocket Propulsion Group | Subsystem Co-Lead | June 2025-present

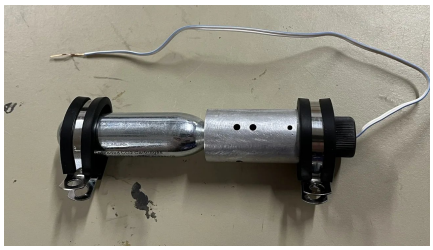
18.7 / 20 lbm
Earlier BP mass / allocation; BP BOM $1,076

823 / 1,230 lbf
Calculated parachute opening / snatch loads

352 L
Operational vacuum chamber; reported test pressure 0.011 atm



Subsystem before CO2 coldflow; historical pretest photo.



Manufactured CO2 cartridge/piercer assembly. Design and fabrication shared by our three-person team.



Door, adapter and seal designed by me; fabricated by my co-lead, who designed the clamps.

Design and Test Leadership

Developed hardware within an inherited dual-break, dual-deploy architecture through four reviews. Compared five published load methods to size lines and check hardware; paired flight computers provide separate firing paths.

Authored the integrated vinyl-charge black-powder SOP and led the second integrated test; collaborated on inspection and failure assessment. One of three charges failed in each test; the second damaged the parachute bag and canopy.

Standalone vinyl-charge successes did not establish integrated reliability. End-seal manufacturing variability is the leading hypothesis; adjacent-charge damage was inferred afterward. Root cause remains unconfirmed.

Designed and sized CO2 piercers using pressure, Lamé thick-wall hoop-stress and thread-shear calculations; contributed CAD, drawings, machining and component testing in a three-person team.

Calculation basis

Recovery lines	Selected 1/2 in, 6,000 lbf Kevlar stock. Design rationale: near-full strength retention, approximately 2× a 50%-efficient knot. Our strength gain remains unmeasured.
CO2 deployment sizing	Calculated 4.66 psi separation requirement for four nylon shear pins plus estimated interface friction. Ideal-gas estimate for the modeled free volume: 10.9 g versus a selected 16 g cartridge.
CO2 piercer housing	Minimum calculated factor of safety (FOS) 2.01 in Lamé hoop stress; internal-thread shear FOS 6.31. Analytical margins, not proof-test results.
Vacuum door / Roark	12 in unsupported diameter, 0.25 in thick 6061-T6 plate; 14.7 psi differential gives 1,663 lbf total load. Recorded bending stress 16.5 ksi; deflection 0.0848 in; minimum yield FOS 2.12 using 240 MPa yield strength. Uniform pressure; simply supported around the outer circumference. Deflection uses the Roark plate model; stress is a separately entered calculator result.
Legacy Catclaw body	Recorded minimum body FOS 2.01 after my material reduction and piston-retention-slot modification. This tie cutter is separate from the CO2 piercer.

Verification Status

Component firing at reported 0.011 atm initial pressure complete. The integrated CO2 trial partially failed at the wiring feedthrough; our team revised the seal using an NPT fitting, epoxy and vacuum putty. Retest and subscale flight testing remain pending. Splice proof load: estimated 1,800 lbf, 1.46 times calculated snatch load, no observed slip; ultimate strength unmeasured. Chamber instrumentation and pressure-data analysis by my co-lead.

Recovery architecture, hardware & tests | CO2 sizing & stress calculations | Chamber design

02 / AERODYNAMICS

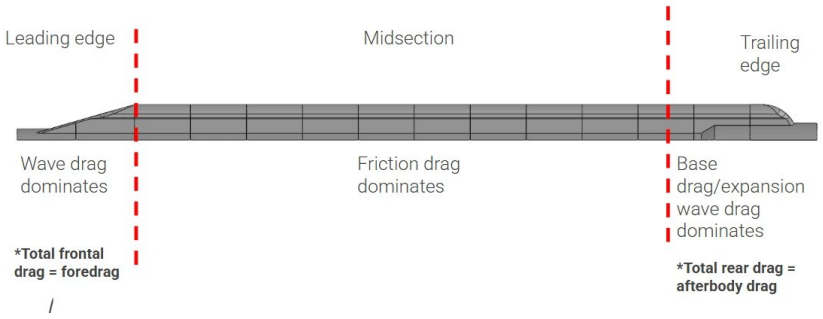
Argo Downcomer Aerocover

Boston University Rocket Propulsion Group | Project Lead | August 2026-present

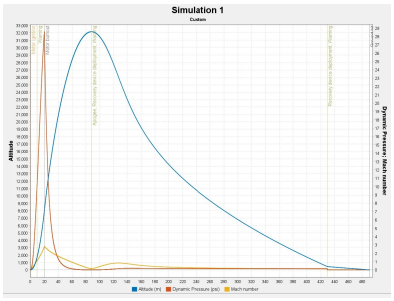
63 in × 1.6 in
Fairing length / nominal height on an 8 in diameter rocket

Mach 2.0-2.81
Priority comparison range from the simulated trajectory

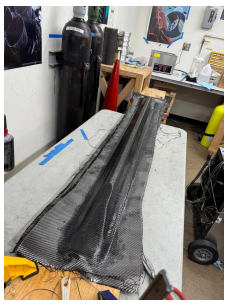
3 concepts
Teammate-owned geometries; shared requirements and analysis by me



My CoDR diagram: a first-principles decomposition, not a measured allocation of drag. Pressure and friction act across the whole fairing.



Trajectory used to select high-dynamic-pressure analysis conditions; simulation, not flight data.



Legacy Icarus fairing: the CFD baseline and a backup flight candidate, not a new design by me.

Analysis

Used D = q Sref CD to prioritize conditions by dynamic pressure as well as Mach. Evaluated leading-edge pressure drag, midsection wetted area and aft pressure recovery together with thermal loads, attachment integrity and manufacturing constraints. Isolated axisymmetric profile rankings cannot establish the performance of a partial fairing attached to the rocket body: junction flow and fin interference require integrated analysis.

Preliminary friction estimate	At Mach 2.81 and recorded Re = 2.04 × 10^8: $C_{D,f} \approx 0.0792$, using 477 in² wetted area, 7.99 in² reference area and a 10% preliminary roughness allowance.
Leading-edge surrogate	A 15° full-cone correlation gives $C_p \approx 0.1888$. It is a screening surrogate for the wedge-like legacy front, not the mounted fairing's total drag coefficient.

Contributions and Status

I developed the common analysis, integration/manufacturing trade and CoDR. Quinn Colleran, Hannah Ruan and Aaron Lin own the three concept geometries. My Mach 2.81 Icarus solve in **Ansys Fluent is converged and under review**; candidate ranking and Ansys adjoint optimization remain pending.

The investigation also led to a nine-profile-family nosecone sweep in RASAero II. A ranking difference from published comparisons prompted further investigation; it does not establish a better flight design.

Concept credits, hand calculations, sources & nosecone sweep | Hand-calculation reference: S. S. Chin, Missile Configuration Design (1961).

03 / BU PLUMMER LABORATORY

Shaker Adapter and Mounting Bolts

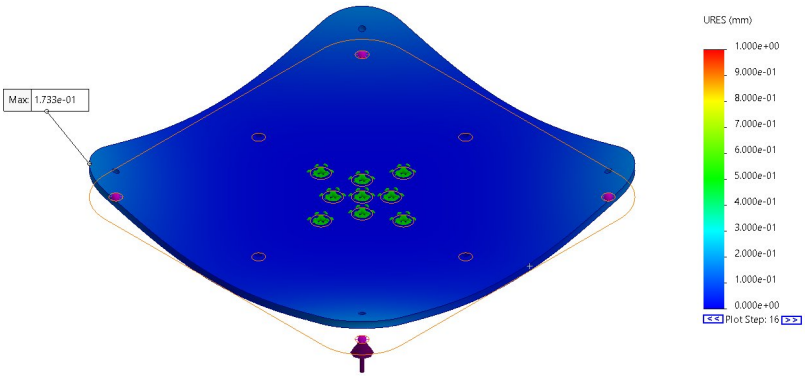
Mechanical Design Assistant | June 2026-present

0.173 mm
Plate displacement predicted in the design-review FEA case

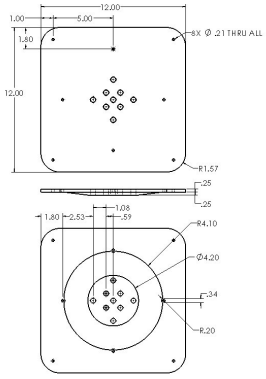
20.1 Hz
Shaker operating frequency; analyzed with a 3 kg fixture load

2.46 FOS
Plate bearing under the modeled combined bolt load

The adapter mounts a physical Ising-model fixture for controlled shaker vibration tests of bistable buckling elements.



SolidWorks plate-displacement result: 0.173 mm maximum. This is a model prediction.



My 12 × 12 in adapter drawing, including underside geometry.

Design and Manufacturing

Replaced an estimated $1,170 outsourced plate/custom-bolt solution with $71.11 of in-house hardware. Developed SolidWorks CAD/FEA, drawings and CAM/G-code; machined the 6061-T6 plate on a Haas VF-2 and checked plate dimensions with calipers against the ±0.005 in drawing tolerance.

Load-path and displacement calculations guided material removal; SolidWorks FEA checked the lightened plate. Finished mass: 1.90 kg from 3.19 kg stock (40.4% stock-to-part reduction).

Drilled 5 mm axial bores in standard medium-carbon-steel M10 bolts for the required shaker interface after unsuccessful stainless-steel machining attempts. The bored bolts and adapter are installed.

Structural calculation basis

Installed joint geometry	M10 × 1.5 with a 5 mm axial bore; modeled remaining tensile area 38.35 mm². Nominal Class 8.8 yield strength 640 MPa; local plate bearing thickness 9.7 mm; thread engagement 9.35 mm.
Per-bolt load path	5 kg allowance (2 kg plate + 3 kg fixture) at 11 G, shared by four bolts: 134.9 N external load per bolt. Workbook conservatively adds the full external load to 16.36 kN preload: 16.50 kN combined. Torque estimate: 21.7 ft-lbf at K = 0.18; joint stiffness sharing is not modeled.
Static FOS by checked mode	Bolt tensile yield: 1.49 under preload plus full external load; preload alone gives 1.50. Plate bearing 2.46; pull-through 4.99; external-thread shear 3.71; internal-thread shear 3.60. These are mode-specific ratios, not an overall dynamic FOS.
Displacement checks	Selected L/240 limit for the 12 in plate span: 1.27 mm; review plate FEA: 0.173 mm. The later 3 kg dynamic fixture model predicts 0.44 mm deformation relative to the base at 20.1 Hz and 13 mm peak-to-peak base motion. These are separate load cases.

Preload maintains joint compression; bolt/joint stiffness determines cyclic load sharing. The 1.49 ratio checks static tensile yield, not fatigue at 20.1 Hz. Preload scatter, separation and fatigue are not quantified here. Displacements are predictions; cost covers plate and bolts only.

Full calculations, drawings, FEA and machining evidence | Sources: my design review and DSRF Calcs; joint interpretation: NASA TM-106943.